3. In the same way, can you guess the next number in the special number sequence given below?

**1, 1, 2, 3, 5, 8, 13, 21, 34,...**

Yes it is 55, how? You have got it by adding 21 and 34. Haven't you?

Are you able to recognize the pattern in the above sequence? Yes, if we add the previous two consecutive terms, we get the next term as

**1+1 =2, 1+2=3, 2+3=5, 3+5=8 , 5+8=13,...**

This special pattern of numbers is called the ***Fibonacci sequence***. Each term in the Fibonacci sequence is called a ***Fibonacci number***.

4. ***Lucas numbers*** form a sequence of numbers like the Fibonacci numbers and they are closely related to the Fibonacci numbers. Instead of starting with 1 and 1, Lucas numbers start with 1 and 3. The Lucas sequence is 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, ... In all the above patterns of numbers, we can see the iterative processes.

**Try these**

i) Find the 10$^{th}$ term of the Fibonacci sequence.

ii) If the 11$^{th}$ term of the Fibonacci sequence is 89 and 13$^{th}$ term is 233 then, what is the 12$^{th}$ term?

E9EYFH

## Fibonacci numbers in nature

We come across the existence of Fibonacci sequence in many natural phenomena like spiral in a shell, arrangement of petals in flowers, branches of a tree, seeds in the head of a sunflower, petals on a daisy, the cells in the bee-hive, etc. Mathematical patterns are found in the distinct marking on animals and the structure of seashells also.












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